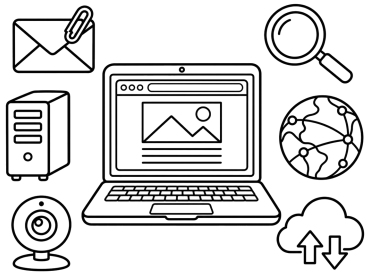
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 3
The Internet and the Electronic Mail



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 3 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Practicals 1 - 4, textbook links and notes, a quiz
- **Writing spaces** - what the students write in each writing space
- **Answers** - chapters 1 to 5, in the same order as the workbook
- **Unit 3 Test** - the marking scheme, common mistakes, homework, a practical test

★ BEFORE THE FIRST LESSON

- ☐ The lab computers can open the web sites of the chapter.
- ☐ Each student has an e-mail account to use in school (Practical 3).
- ☐ You have the writing-space notes of each chapter ready to read out.

"Students remember the Internet best when they see it work: let them open the sites, send the mails and follow the steps."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Label the URL	5
	B Correct IP address?	6
	C Find the domain names	8
	D True or false	6
Chapter 2	A Put the steps in order	6
	B Be the DNS server	4
	C Which protocol?	5
	D Downloading or uploading?	6
Chapter 3	A True or false	6
	B Which service is it?	7
	C Web browser or search engine?	8
	D Word search	10
Chapter 4	A Parts of e-mail addresses	8
	B Label the e-mail window	6
	C Who can see the addresses?	6
	D Which folder?	5
Chapter 5	A Match A and B	8
	B Which way to communicate?	6
	C IaaS, PaaS or SaaS?	6
	D Answer these	4
End	Unit 3 Test	40
TOTAL		**166**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Before the unit, open every web site of the chapter in the lab yourself: moe.gov.lk, nie.lk, doenets.lk and edupub.gov.lk.
- Write a long URL on the board. Students come up and draw a bracket under each part and name it.
- Let pairs swap books and check each other's tick tables before you mark them.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (25)	Ch 2 (21)	Ch 3 (31)	Ch 4 (25)	Ch 5 (24)	Test (40)	Total (166)
1								
2								
3								
4								
5								
6								
7								
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TEACHER GUIDE

Practicals at a Glance

Practicals 1 to 4 - Practical 4 is the textbook Activity on page 80

Practical	Students do	Check that ...
1 (pp. 65 - 67)	Visit the education web sites; find the IP address of the computer	they type each URL in the address bar, name the protocol, www, domain name, path and resource, and write a correct IP address (four values, 0 - 255); nothing is changed.
2 (pp. 70, 73)	Browse a web site; search with a search engine	they find the address bar, the home page and the hyperlinks of doenets.lk, search with key words and open a hyperlink of the results.
3 (pp. 74 - 76)	Send an e-mail with Cc, Bcc and an attachment; look at the folders	the mail has To, Cc, Bcc, a subject, a polite message and a small attachment; they find it in Sent, know Inbox, Drafts, Trash and Spam, and sign out.
4 (Activity, p. 80)	Find the items of the textbook Activity on a computer	they show a web browser, a URL and its domain name, a search engine, an e-mail address and an IP address, then match the columns in the exercise Match A and B.

! SAFETY AND MATERIALS

- Check the Internet connection and the school filter before each practical; open every web site yourself.
- Practical 1: Windows 10 shows the IPv4 address under Settings, Network & Internet, Properties. Students only look - no setting is changed.
- Practical 3: school e-mail accounts, or accounts made with the parents' permission. Students never share a password and always sign out.
- Section 3.18: if you can, call another class or school with a web cam and communication software.

Textbook links

Workbook	Textbook pages	Practicals
Chapter 1 - The Internet and its addresses	64 - 67	1
Chapter 2 - How the Internet works	67 - 69, 73	-
Chapter 3 - Services of the Internet	69 - 73	2
Chapter 4 - Electronic mail	74 - 76	3
Chapter 5 - Communication and the cloud	77 - 80	4 (Activity)
Unit 3 in One Page	80 (summary)	-

★ TEACHING TIPS

- Write five IP addresses on the board, some of them wrong. Students check: four values, each from 0 to 255.
- Use the word list as a quick spelling test at the start of a lesson.

! TEXTBOOK NOTES - THE WORKBOOK IS CORRECT

- Figure 3.2 (p. 66) is captioned "Using IF addresses": read **IP addresses**.
- The country domain table (p. 67) has "Japan | jp" and "United States | us" in the wrong columns: the domains are **jp** (Japan) and **us** (United States).
- The table on p. 66 gives gov = "Public", but the text says gov represents **the government**: the workbook writes Government (public). Accept both. For net the textbook gives "Web" (accept network too).
- Heading 3.1.5 "How the internet work" means works; Figure 3.14 "Member Massage" means **Member message**; p. 79 "Interner connections" means Internet connection; p. 74 "identity this person" means identify.
- Activity (p. 80): "syallabus" is **syllabus**. In the second table, "Web address" (for Mozilla Firefox) must be **Web browser**, and "Domain name" (converts domain name into IP addresses) must be the **Domain Name Server**. The workbook exercise Match A and B uses the corrected words.
- P. 65 calls the WWW "the main server"; p. 70 calls it a **service** of the Internet, as the workbook does. The workbook writes Sir Tim Berners-Lee with a hyphen.
- The workbook numbers its sections 3.1 - 3.20 in its own order. The practical steps are written from the textbook pages for Windows 10.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What is the Internet?	**A collection of computer networks around the world**
2. Who looks after the protocols of the Internet?	**The Internet Society**
3. How many values has an IP address?	**Four, each from 0 to 255**
4. Who gives IP addresses?	**An Internet Service Provider (ISP)**
5. What is the Top Level Domain of nie.lk?	**lk**
6. What does the DNS do?	**Translates a domain name into an IP address**
7. Which protocol is used for e-mail exchange?	**SMTP**
8. Who is the father of the WWW?	**Sir Tim Berners-Lee**
9. In exams@doenets.lk, what is exams?	**The user name**
10. Which folder keeps unwanted e-mail?	**Spam (Junk)**

★ TEACHING TIPS

- Play "Be the DNS": one student holds a card of domain names and IP addresses; the class asks for a name and gets the number back.
- Act out Figure 3.3 with six students: the user, the resolver, the DNS server, the browser, the web server and the screen.
- Show the DNS at work on the teacher computer: in the Command Prompt, nslookup www.google.lk gives back an IP address.
- For the word search, tell weaker students the first letter of each word.

WRITING SPACES

What Students Write

Read each note out or write it on the board - the students copy it into the space

Chapter 1

No.	Page	What the students write
1	4	**Parts of a URL** http:// = protocol, www. = World Wide Web, edupub.gov.lk = domain name, /e-books/english/ = path, ict.pdf = resource.
2	4	**IP addresses** An IP address identifies a computer in the Internet: four values from 0 to 255 with dots. An ISP gives it.

Chapter 2

No.	Page	What the students write
3	7	**The Domain Name System** The DNS translates a domain name into an IP address: www.yahoo.com = 209.191.122.70.
4	7	**How a web page opens** The user enters the URL; the resolver gets the IP address from the DNS server; the browser sends an http get command to the web server; the server returns the HTML data stream; the browser renders the page.
5	8	**Protocols** HTTP - HTML documents, TCP/IP - IP addresses, FTP - files, SMTP - e-mail, ICMP - error messages.

Chapter 3

No.	Page	What the students write
6	11	**The World Wide Web** A large collection of electronic documents. Father: Sir Tim Berners-Lee. Opened with a web browser.
7	12	**Search engines** Type the key words, click the search button, click the hyperlinks. E.g. google.lk, yahoo.com.

Chapter 4

No.	Page	What the students write
8	14	**E-mail addresses** exams@doenets.lk: exams = user name, doenets.lk = domain name, @ separates them.
9	15	**To, Cc and Bcc** Cc = carbon copy. Bcc = blind carbon copy: the To and Cc receivers cannot see the Bcc address.

TIP: Keep a poster of the five protocols with their full names and uses on the wall during the unit.

Chapter 5

No.	Page	What the students write
10	18	**Video conference** Needs computers, a web cam, an Internet connection and communication software (audiovisual).
11	19	**Cloud computing** IaaS - servers and storage; PaaS - a platform to develop software; SaaS - software without installing.

ANSWERS

Chapter 1

The Internet and its addresses

A Label the URL

5 marks

1. hypertext transfer protocol	**2.** World Wide Web	**3.** domain name
4. path where the resource is located	**5.** resource	

1 mark each. The URL is the textbook Activity URL http://www.nie.lk/syllabus/ol/ict.pdf.

B Correct IP address?

6 marks

IP address	Correct	Not correct
172.64.85.42	☑	☐
193.213.78.256	☐	☑
209.191.122.70	☑	☐
100.1.0	☐	☑
74.125.20.103	☑	☐
192.168.2.10.1	☐	☑

Not correct: 2: 256 is more than 255. 4: only three values. 6: five values.

C Find the domain names

8 marks

URL	Domain name	Top Level Domain
http://www.moe.gov.lk	**moe.gov.lk**	**lk**
http://www.yahoo.com	**yahoo.com**	**com**
http://www.doenets.lk	**doenets.lk**	**lk**
http://www.pinterest.com	**pinterest.com**	**com**

1 mark for each domain name and 1 mark for each Top Level Domain.

D True or false

6 marks

1. F	2. T	3. T	4. F	5. T	6. F

False: 1 (the Internet has no single owner), 4 (lk is the Top Level Domain), 6 (a URL must have the parts from HTTP to the domain name).

ANSWERS

Chapter 2

How the Internet works

A Put the steps in order

6 marks

The numbers in the boxes, from the top down (1 mark each):

a. 4	b. 1	c. 6	d. 3	e. 5	f. 2

B Be the DNS server

4 marks

The DNS server's table

Domain name	IP address
www.volusion.com	69.49.191.216
www.google.com	74.125.20.103
www.facebook.com	173.252.110.27
www.yahoo.com	209.191.122.70

The browser asks for	The DNS server replies
www.google.com	**74.125.20.103**
www.yahoo.com	**209.191.122.70**
www.facebook.com	**173.252.110.27**
www.volusion.com	69.49.191.216

1 mark each. The last row is the reverse look-up: the name of 69.49.191.216.

C Which protocol?

5 marks

1. HTTP	2. SMTP	3. FTP	4. ICMP	5. TCP/IP

D Downloading or uploading?

6 marks

What happens	Downloading	Uploading
Nimal saves an e-book from edupub.gov.lk on his computer.	☑	☐
Kavindi sends her project file to the school server.	☐	☑
A song from a web site is saved on a phone.	☑	☐
A teacher puts the marks sheet on the school web server.	☐	☑
You get the O/L ICT syllabus file from nie.lk.	☑	☐
A file goes from a client computer to an FTP server.	☐	☑

ANSWERS

Chapter 3

Services of the Internet

A True or false

6 marks

1. F	**2. F**	**3. T**	**4. T**	**5. F**	**6. T**

False: 1 (a web site has several web pages), 2 (large files cannot be sent as attachments - use FTP), 5 (the media server bypasses the web server).

B Which service is it?

7 marks

1. World Wide Web	**2.** E-mail	**3.** FTP	**4.** Remote access
5. File sharing	**6.** Streaming of media	**7.** Search engines	

C Web browser or search engine?

8 marks

Name	Web browser	Search engine
Google Chrome	☑	☐
www.google.lk	☐	☑
Mozilla Firefox	☑	☐
www.yahoo.com	☐	☑

Name	Web browser	Search engine
Safari	☑	☐
www.ask.com	☐	☑
Opera	☑	☐
www.msn.com	☐	☑

Google Chrome is a web browser; www.google.lk is a search engine.

D Word search

10 marks

		H	O	M	E	P	A	G	E	
G	N	I	R	A	H	S		H		
						D		A	H	S
H	C	R	A	E	S	R		C	Y	T
	W					O		K	P	R
		E				W		I	E	E
			B			Y		N	R	A
				S		E		G	L	M
					I	K			I	I
		R	E	M	O	T	E		N	N
		B	R	O	W	S	E	R	K	G

Word	Starts at	Direction
BROWSER	row 11, col 3	across
HYPERLINK	row 3, col 10	down
WEBSITE	row 5, col 2	slanting down
HOMEPAGE	row 1, col 3	across
STREAMING	row 3, col 11	down
SEARCH	row 4, col 6	across, backwards
KEYWORDS	row 9, col 7	up
REMOTE	row 10, col 3	across
HACKING	row 2, col 9	down
SHARING	row 2, col 7	across, backwards

ANSWERS

Chapter 4

Electronic mail

A Parts of e-mail addresses

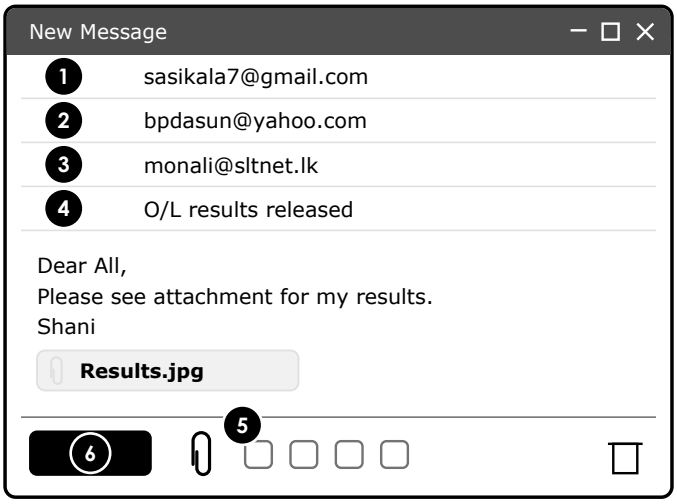
8 marks

E-mail address	User name	Domain name
sasikala7@gmail.com	**sasikala7**	**gmail.com**
bpdasun@yahoo.com	**bpdasun**	**yahoo.com**
monali@sltnet.lk	**monali**	**sltnet.lk**
somasiri@gmail.com	**somasiri**	**gmail.com**

1 mark for each user name and 1 mark for each domain name.

B Label the e-mail window

6 marks



1. **To**
2. **Cc**
3. **Bcc**
4. **Subject**
5. **Attachment**
6. **Send**

C Who can see the addresses?

6 marks

Question	Yes	No
Can Sasikala (To) see Dasun's (Cc) address?	☑	☐
Can Dasun (Cc) see Monali's (Bcc) address?	☐	☑
Can Monali (Bcc) see Sasikala's (To) address?	☑	☐
Can Sasikala (To) see Monali's (Bcc) address?	☐	☑
Can Monali (Bcc) see Dasun's (Cc) address?	☑	☐
Do all three friends get the e-mail?	☑	☐

The To and Cc receivers can see each other; only the Bcc receiver sees all the addresses.

TIP: Say it often: a web browser opens web sites; a search engine is a web site that finds other web sites.

D Which folder?

5 marks

1. Inbox	**2.** Drafts	**3.** Sent	**4.** Trash	**5.** Spam

ANSWERS

Chapter 5

Communication and the cloud

A Match A and B

8 marks

1. E-mail address → **C**	**2.** Domain name → **E**
3. Uniform Resource Locator → **G**	**4.** IP address → **A**
5. Web browser → **H**	**6.** A service of the Internet → **F**
7. Domain Name Server → **B**	**8.** Search engine → **D**

1 mark each. The textbook prints "Web address" for 5 and "Domain name" for 7 (see the textbook notes).

B Which way to communicate?

6 marks

What people do	E-mail	Instant message	Video conference	Social network
Two friends chat with short messages very fast.	☐	☑	☐	☐
Doctors in two countries talk with web cams.	☐	☐	☑	☐
Kamal shares his trip photos with his relatives.	☐	☐	☐	☑
A school sends a letter and a file to a friend school.	☑	☐	☐	☐
Principals in five districts meet with audio and video.	☐	☐	☑	☐
Nimali follows her old classmates' daily activities.	☐	☐	☐	☑

C IaaS, PaaS or SaaS?

6 marks

In the cloud ...	IaaS	PaaS	SaaS
A company uses server space and a firewall from the cloud.	☑	☐	☐
Software is developed in the cloud without compilers.	☐	☑	☐
A word processor is used in a browser, not installed.	☐	☐	☑
Data centres give space to store data and software.	☑	☐	☐
An OS, a database and a web server for developers.	☐	☑	☐
A spreadsheet is used in a browser on an Android phone.	☐	☐	☑

D Answer these

4 marks

1. Write the four requirements of a video conference. (2 marks)

Computers, a web cam, an Internet connection and communication software with audiovisual facilities (two correct = 1 mark, all four = 2 marks).

2. Write one good result and one problem of social networks. (2 marks)

Good: social relationships are widened (people share details, photos, videos and daily activities) (1 mark). Problem: they can create social issues (1 mark).

MARKING SCHEME

Unit 3 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (a) 192.168.2.10	**2.** (d) syllabus.asp	**3.** (c) SMTP	**4.** (c) DNS server
5. (b) downloading	**6.** (c) Mozilla Firefox	**7.** (c) Bcc	**8.** (a) A and B only

2 Fill in the blanks

8 marks (1 each)

1. Society	**2.** 255	**3.** Level	**4.** HTTP
5. FTP	**6.** home	**7.** domain	**8.** spam

3 Look at the URL and the e-mail address

8 marks (1 each)

1. hypertext transfer protocol (HTTP)	**2.** World Wide Web (WWW)
3. domain name	**4.** path where the resource is located
5. resource	**6.** user name
7. domain name	**8.** org

URL http://www.unicode.org/consortium/consort.html; e-mail address monali@sltnet.lk. Accept "protocol" for 1 and "path" for 4.

TEACHING TIPS

- For Practical 3 use the school e-mail system, or accounts made with the parents' permission.
- Teach polite e-mails: a clear subject, a greeting, short sentences and the sender's name at the end.
- Show three good e-mails from Practical 3 (with permission) to the class as models.

4 Answer these

16 marks

1. What is an IP address? Who gives IP addresses? (2 marks)

An address that identifies each computer in the Internet individually: four values from 0 to 255 separated by dots (1 mark). An Internet Service Provider (ISP) (1 mark).

2. Explain how the DNS helps a web browser to open www.yahoo.com. (3 marks)

The request is first sent to the DNS server (1 mark). It translates www.yahoo.com into the IP address 209.191.122.70 (1 mark). The real connection to the web site is made with this IP address (1 mark).

3. Name three Internet servers and write what each one does. (3 marks)

Web server - stores web pages and provides them to the clients (1 mark). Mail server - stores e-mail and provides it to the clients (1 mark). DNS server - translates URLs (domain names) to IP addresses (1 mark).

4. Write the four requirements of a video conference. (2 marks)

Computers, a web cam, an Internet connection, communication software with audiovisual facilities (two correct = 1 mark, all four = 2 marks).

5. Shani e-mails To Sasikala, Cc Dasun, Bcc Monali. (a) Who cannot see Monali's address? (b) How is a file sent with it? (c) Which folder keeps the sent mail? (3 marks)

(a) Sasikala and Dasun - the To and Cc receivers (1 mark). (b) She attaches it with the attachment (paper clip) button (1 mark). (c) In the Sent folder (1 mark).

6. What is Software as a Service (SaaS)? Give one example. (3 marks)

Software installed in the cloud that is used without installing it on your device (1 mark); it reduces the cost and complexity of buying and managing hardware and software (1 mark). E.g. word processing or spreadsheet software used through a web browser on an Android device (1 mark).

TEACHING TIPS

- Talk about safety: never share a password, never open unknown attachments, think before you post.
- If you can, hold a short video conference with another school, so students see the four requirements work.
- Use cloud services students already know (e-mail in a browser, online documents) as SaaS examples.
- Read each writing-space note slowly, or write it on the board, and check that students copy it neatly.
- Mark the unit test together the next day. Students learn most from the questions they got wrong.
- Ask students to explain each new word in one short sentence: short, exact answers win O/L marks.

WHAT STUDENTS USUALLY GET WRONG

- They write three or five values in an IP address, or a value above 255. It has **four** values, each **0 - 255**.
- They call www the domain name. In http://www.edupub.gov.lk/... the domain name is **edupub.gov.lk**.
- They give gov as the Top Level Domain of edupub.gov.lk. The TLD is the **right-most** domain: lk.
- They mix up a **web browser** (Mozilla Firefox, Google Chrome) and a **search engine** (google.lk, yahoo.com).
- They mix up **downloading** (server to client) and **uploading** (client to server).
- They think the To and Cc receivers can see the Bcc address. Only the **Bcc receiver** sees all.
- They mix up the cloud services: **SaaS** = software used without installing it; **PaaS** = a platform to develop software; **IaaS** = servers and storage.

HOMEWORK AND EXTRA PRACTICE

- Write the URLs of five Sri Lankan web sites. Underline each domain name and circle its Top Level Domain.
- Draw the six steps of Figure 3.3 from memory and explain them to someone at home.
- Make a table of the five protocols with their full names and their uses.
- Write a polite e-mail to your class teacher on paper: To, Cc, Subject, the message and an attachment name.
- Fast learners: which instant message tools does your family use? Which of them can make video calls?

PRACTICAL TEST - 20 MINUTES, 20 MARKS (OPTIONAL)

In the lab, one student at a time, after the unit. Read the steps aloud. Mark on the screen.

Step	The student must ...	Marks
1	Open http://www.nie.lk and name the parts of its URL.	3
2	Show the IPv4 address of the computer and say why it is a correct IP address.	2
3	Search for "Department of Examinations" with a search engine and open its web site.	3
4	Point to the address bar, a hyperlink and the home page of that web site.	3
5	Sign in to the e-mail account (the student types the password).	1
6	Send an e-mail with To, Cc, a subject, a message and an attachment.	5
7	Find the mail in the Sent folder, then sign out.	3
	Total	**20**